

Classic Six Pack Gauges on a 4-inch LCD Screen Running as a Mobiflight Community Device for Flight Simulation (Build Guide)



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Introduction

I started this project to build my own six-pack gauges for flight simulation, but I used a 4-inch LCD screen instead of 3D-printed physical parts and stepper motors because I don't like the noise they generate.

The display is a 4-inch LCD display with an integrated ESP32 S3 microcontroller. It runs as a Mobiflight (<https://www.mobiflight.com/en/index.html>) custom (aka community) device to integrated with flight simulators such as X Plane and MSFS.

Features

- Each gauge runs on its own display
- Relatively low cost, about 30-40 USD per gauge

Pre-requisites

- Knowledge/experience with Mobiflight (<https://www.mobiflight.com/en/index.html>) is required, as this runs as a Mobiflight community device
- You should be able to configure additional Mobiflight boards and add encoders to control things such as the True Airspeed (TAS) knob in the Airspeed Indicator, Heading Knob in the Heading Indicator, Baro Knob in the Altimeter. Encoder configuration in Mobiflight is beyond the scope of this document. Please see here for detailed instructions on Mobiflight configuration: <https://docs.mobiflight.com/>

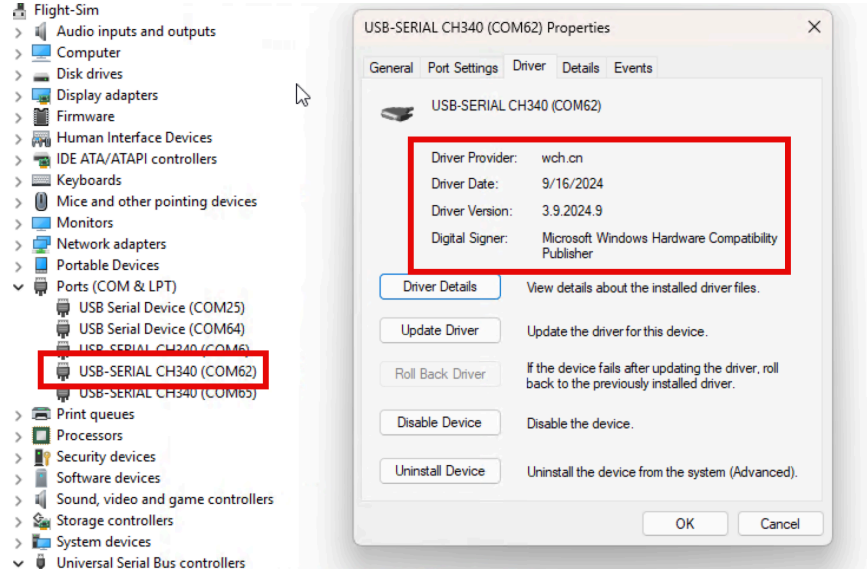
Bill of Materials

- Guition Esp32-s3-4848s040 4-inch LCD screen
 - <https://www.aliexpress.com/item/3256808028364930.html>
 - **Hint!:** Search around AliExpress for deals! The price can range from about 18 USD to 30 USD, and sometimes lower if there are sales events
- EC11 encoders to be used for knobs to control the TAS knob, Heading Knob, etc.
- Additional Mobiflight-supported boards (Arduino Nano/Uno, RP2040, etc.) to connect the encoders. For details, see here: <https://docs.mobiflight.com/boards/>.
- USB Type C cables
- USB Hub to connect the devices to your gaming PC
- 3D printer and filaments if you want to print the case
- Wires, soldering materials, etc., to solder the encoders to your Mobiflight board

Preparations

- Install the CH340G driver
 - Download the driver from here: https://www.wch-ic.com/downloads/CH341SER_ZIP.html
 - Extract the zip file and run as administrator the "SETUP.exe"
 - When you connect your Guition display, open the Windows Device Manager, expand "Ports (COM & LPT)", right-click the COM port for your device, and choose

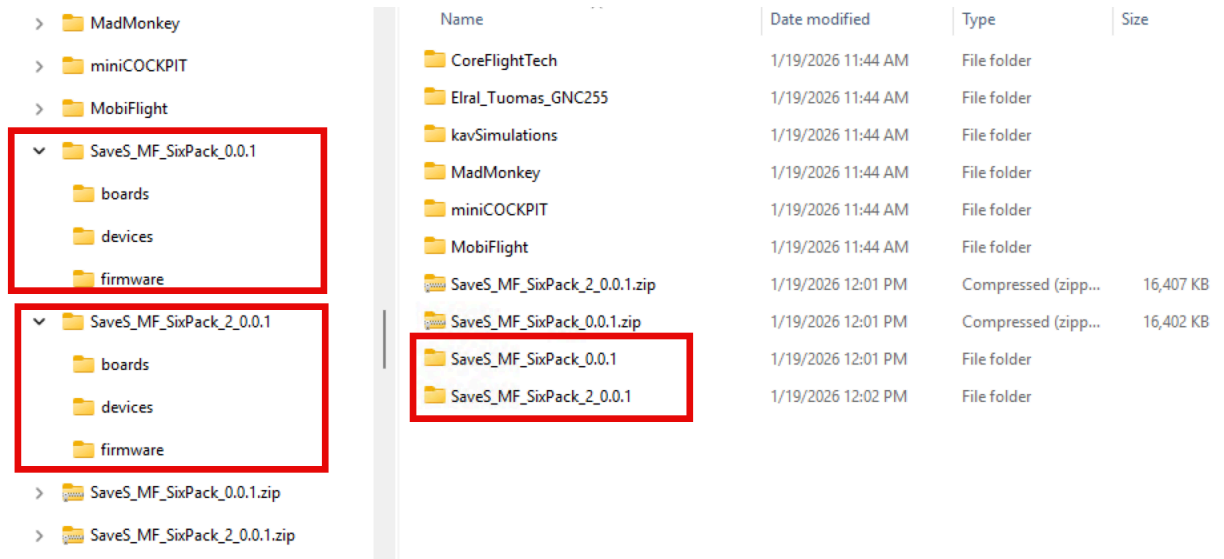
“Properties”. In the “Properties” pop-up, go to the “Drivers” tab and check that the driver date is at least “9/16/2024”. Please see the screenshot below.



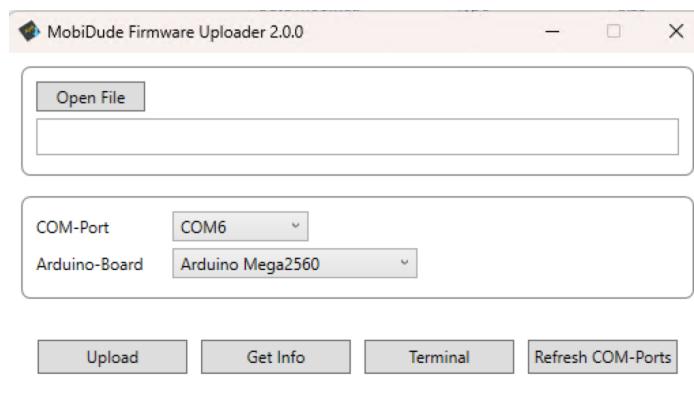
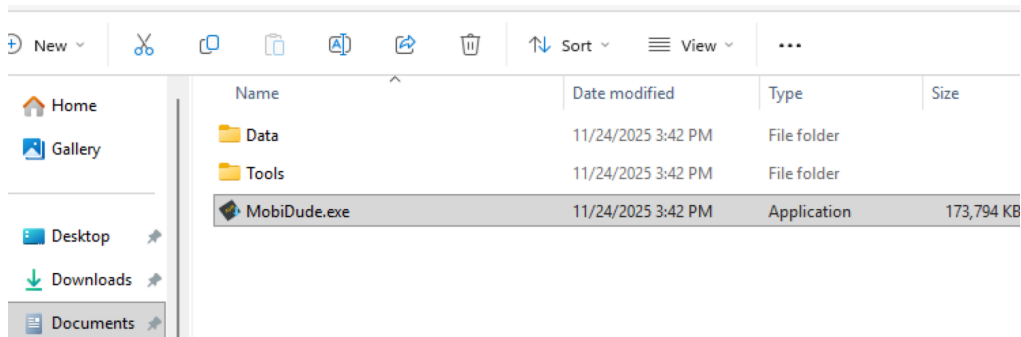
- Download and install Mobiflight (if not already installed) from here: <https://www.mobiflight.com/en/index.html>. In this guide, I will use the officially released version of Mobiflight (not the Beta version!).
- Download and extract the Mobidude tool from elral's GitHub repository. Please make sure that you have version 2.0.0 (later versions have issues!). This tool is used to flash firmware to the Guition LCD screen, as Mobiflight provides no official support for it.
 - <https://github.com/elral/MobiDude/releases/tag/2.0.0>
- Download the Mobiflight Six Pack Gauges firmware from the “firmware” folders from the links below. There are 2 firmwares:
 - The link below contains the firmware for the Airspeed Indicator, Altimeter, Turn Coordinator, and Heading Indicator gauges. The firmware file is called “SaveS_MF_SixPack_0.0.1.zip”
 - https://github.com/savesabanal01/MF_SixPack_480x480_1/tree/MF_SixPack
 - The link below contains the firmware for the Vertical Speed Indicator and the Attitude Indicator gauges. The firmware file is called “SaveS_MF_SixPack_2_0.0.1.zip”

Flashing the Firmware

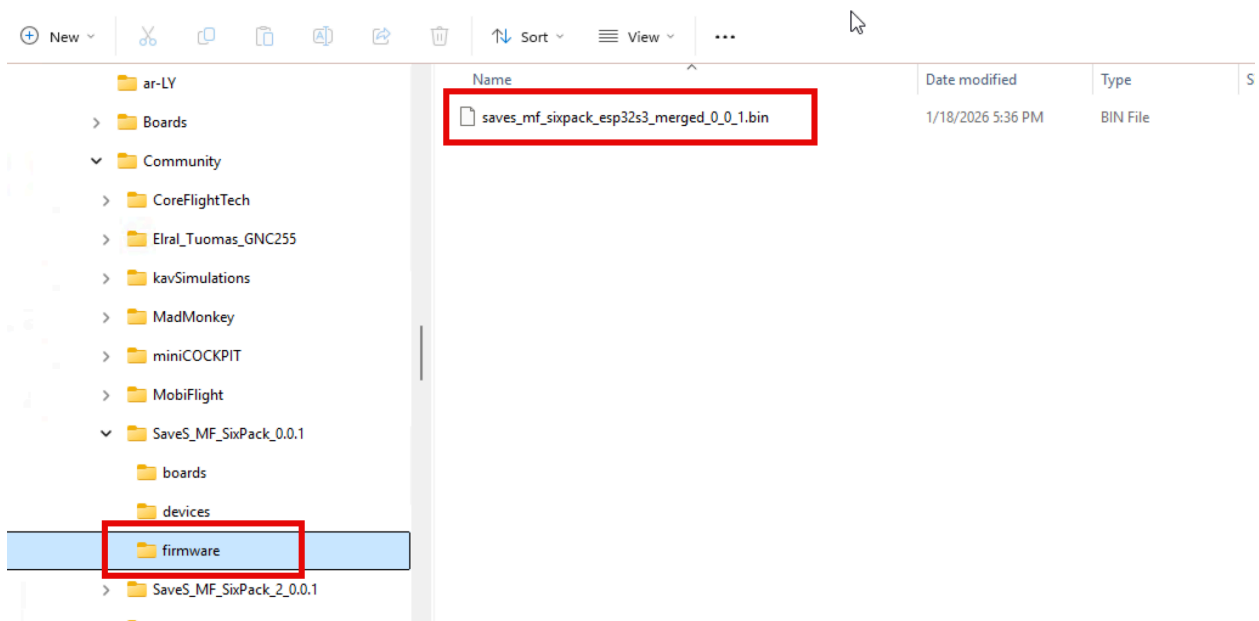
- Open your Windows Explorer, using Win+E, and paste “%LocalAppData%\MobiFlight\Mobiflight Connector\Community” (without the quotation marks). Alternatively, you can navigate to “C:\Users\<your username>\AppData\Local\MobiFlight\MobiFlight Connector\Community”. Please note that you may need to enable the “hidden files” option in Windows Explorer
- Copy the “SaveS_MF_SixPack_0.0.1.zip” and “SaveS_MF_SixPack_2_0.0.1.zip” firmware to the above folder and extract them. The folder structure should look like this:



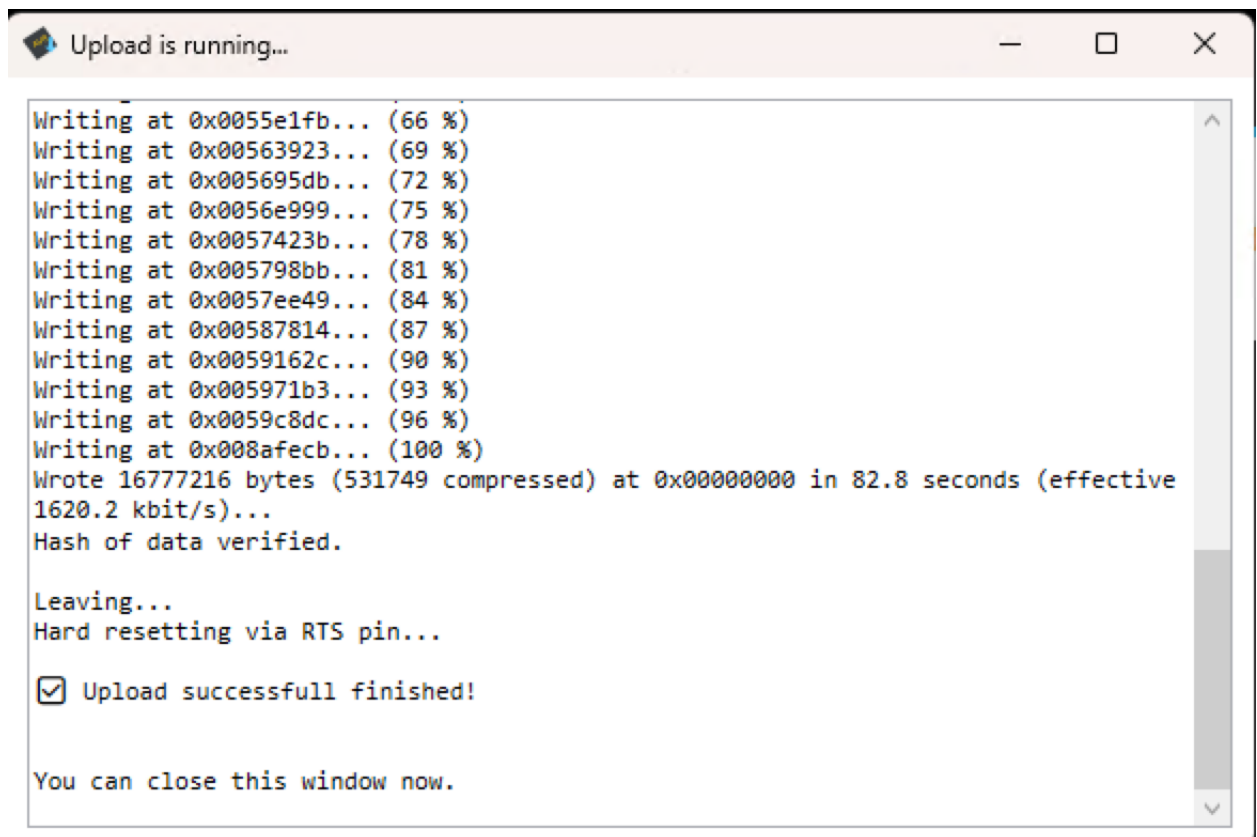
- Connect your Guition LCD screen. Please note the correct COM port (if needed, open the Windows Device Manager, disconnect and connect your Guition LCD screen to see which port it is connected to)
- Go to the folder where you extracted the “MobiDude” tool and double-click on “MobiDude.exe”



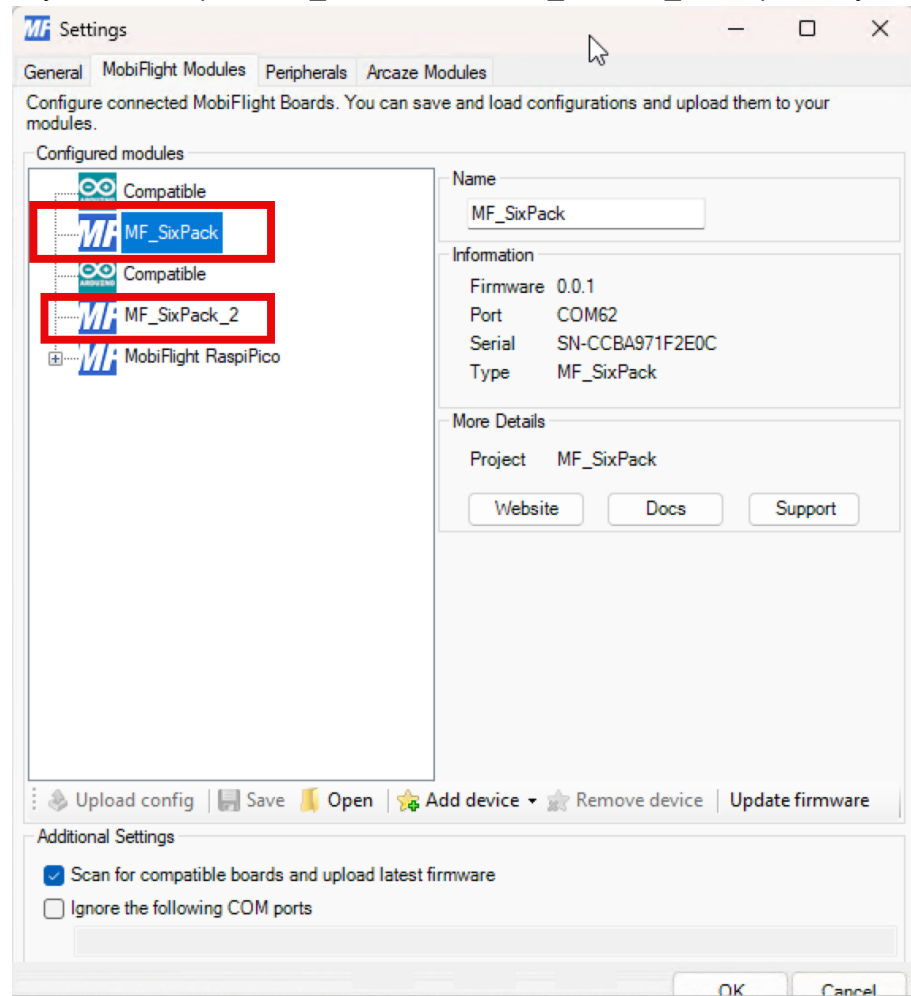
- Click on “Open File” of the MobiDude tool and choose the firmware file. It should be in the “firmware” folder when you extract the respective firmware zip files. See example below. The firmware files are “saves_mf_sixpack_esp32s3_merged_0_0_1.bin” and “saves_mf_sixpack_2_esp32s3_merged_0_0_1.bin” respectively.



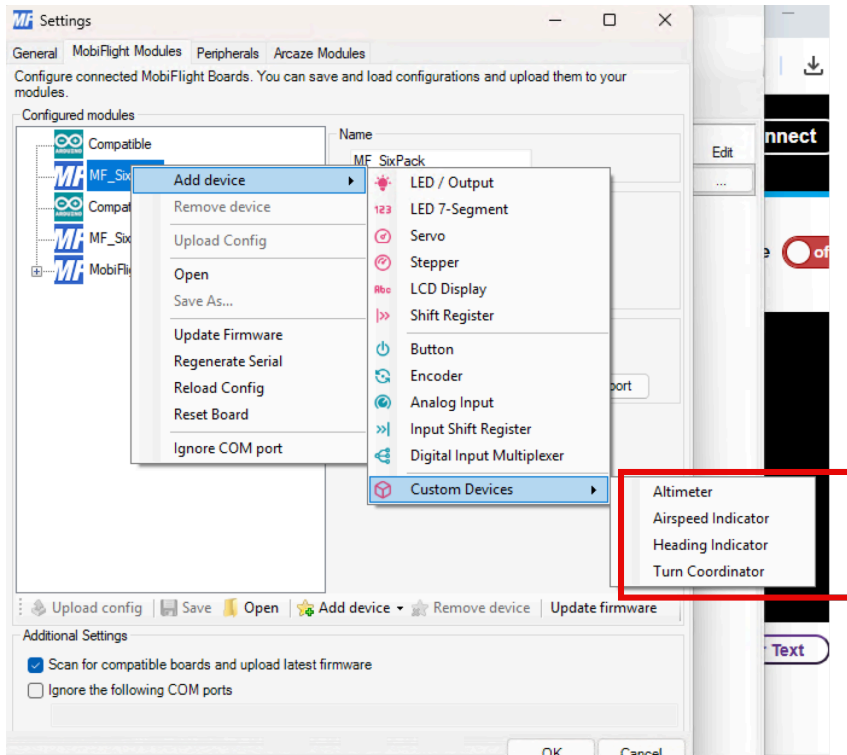
- In the “MobiDude” tool screen, choose the correct COM port (in my case, COM62), and choose “ESP32-S2/S3” in the “Arduino Board” drop-down, and click “Upload”.
- Wait for the “MobiDude” to complete the firmware flash (it will take a few minutes) and close the window.



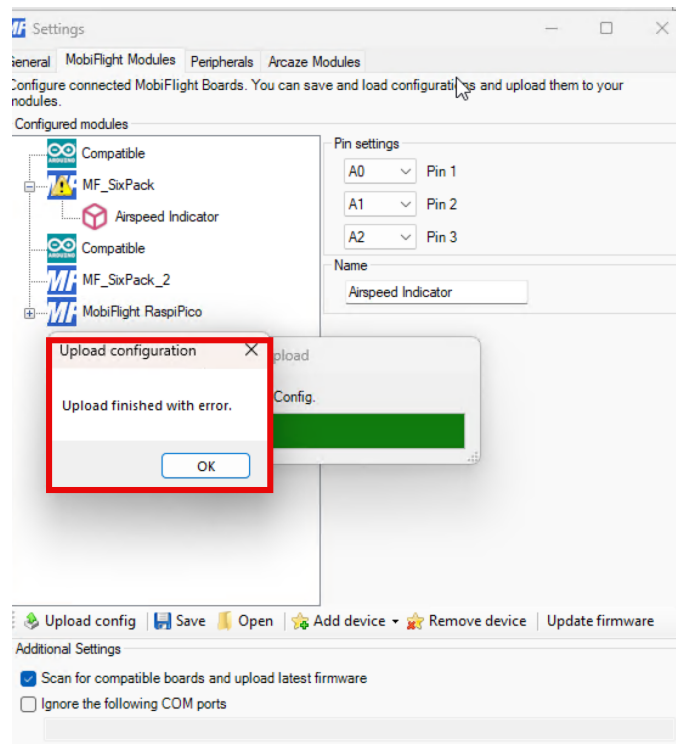
- Open Mobiflight and open Mobiflight Modules. Note: When you run Mobiflight, you might get a “New Firmware Available” pop-up, but you can ignore this. In my example screenshot below, I have 2 Guition LCD screens connected and I flash the 1 firmware on each. They will show up as “MF_SixPack” and “MF_SixPack_2”, respectively.



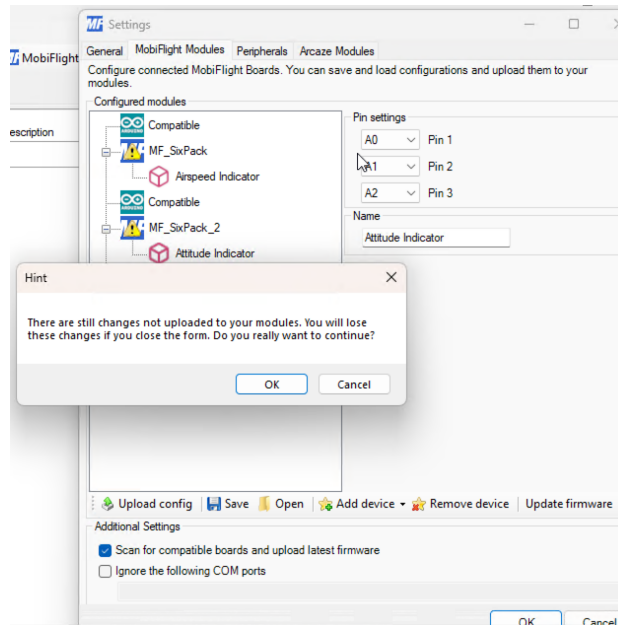
- Right-click on the boards (“MF_SixPack” in the example below), go to “Add device” -> “Custom Devices”. For “MF_SixPack”, you can choose between the Altimeter, Airspeed Indicator, Turn Coordinator and Heading Indicator. For MF_SixPack_2, you can choose between Attitude Indicator and Vertical Speed Indicator.



- Once you have chosen which gauge you want (e.g. Altimeter), click on “Upload config”. Note: In some cases, you might get an “Upload finished with error” pop-up, but this is a known issue and should be fine.

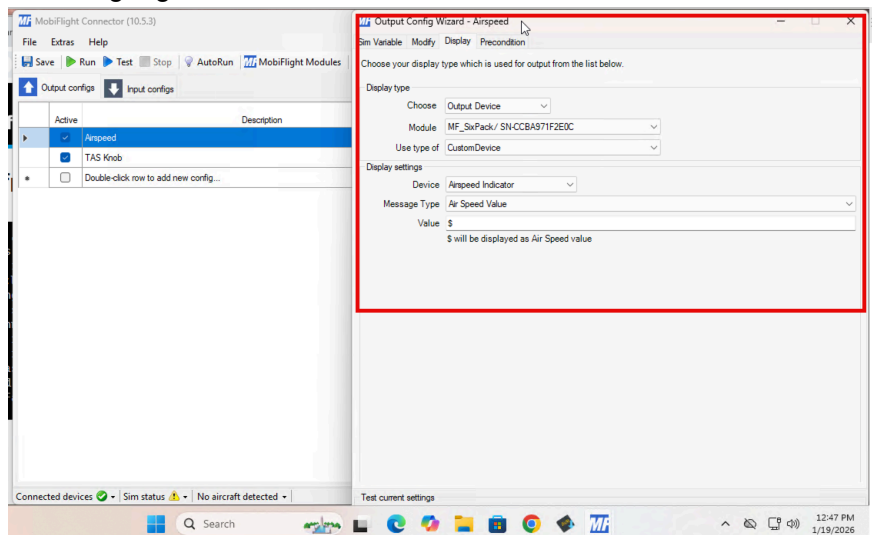


- Close the Mobiflight settings pop-up. You will get a pop-up that changes are not upload, but this is a known issue, and you can ignore it.



Configuring the Custom Devices

- Once the Mobiflight custom firmware is flashed and the configs are uploaded, they can be configured as Mobiflight output configs. For example, for the Airspeed Indicator, you can add the values for the airspeed and the True Airspeed Knob, so that the custom firmware can use them to render the gauges.



- I have also included some Mobiflight Config files for X Plane (as I don't have MSFS). But finding the right sim variables for MSFS using Mobiflight should be straightforward. The config files are under the folder "Mobiflight_Configs":
 - https://github.com/savesabanal01/MF_SixPack_480x480_1/tree/MF_SixPack/Mobiflight_Configs

- https://github.com/savesabanal01/MF_SixPack_480x480_2/tree/MF_SixPack_2/Mobiflight_Configs
- **Note!** Also use another Mobiflight device (Arduino Nano/Uno, RP2040) to connect the encoders to control the Airspeed Indicator TAS Knobs, Altimeter Baro Knob, etc.!!

Guition Case and Stand

- For convenience, I also include a 3D printed case and stand design for the Guition LCD Display under the “Guition Case” folder.
https://github.com/savesabanal01/MF_SixPack_480x480_1/tree/MF_SixPack/Guition%20Case

Support

- For issues, please join the Mobiflight Discord Server and open a thread under “custom-devices-“ so we can support you better.
 - Mobiflight Discord invite: <https://discord.com/invite/U28QeEjBV>

Mobiflight Rocks! And most of all, have fun, take it easy, and enjoy!